

3.1 Functions of Two or More Variables

Definition: A function f of two variables is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on, that is, $\{f(x, y) \mid (x, y) \in D\}$.

Note: Familiar operations like addition can be written as a two variable function. For example, the function $f(x, y) = x + y$ is the function which spits out the sum of two numbers. Using our mapping notation we can write $f: \mathbb{R}^2 \rightarrow \mathbb{R}$.

Example 1: Consider $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$.

a) Find $f(2, 1)$.

b) Write the domain of $f(x, y)$ in set notation.

c) Sketch a picture of the domain of f .

Example 2: Write the domain and range of the following function in set notation: $f(x, y) = \sqrt{4 - x^2 - y^2}$.

Exploration:

From the above example we see that we can think of $f(x, y)$ as the z variable. Thus, we can graph a function of two variables in $\mathbb{R}^3$. Moreover, the graph of a two variable function is a surface in $\mathbb{R}^3$.

Definition: The level curves of a function f of two variables are curves with equations $f(x, y) = k$, where k is a constant in the range of f .

Note: Level curves are closely related to trace curves, and are very helpful when sketching a graph of a two variable function by hand.

Example 3: Consider the function $f(x, y) = x^2 + y^2 - 2$.

a) State the domain and range of this function.

b) Sketch the level curves for the function for the values $k = -2, -1, 0, 2$.

c) Sketch a graph of $f(x, y)$.

Note: We recognize the function in this example as having a graph which is an elliptic paraboloid.

Note: Two dimensional graphs of level curves (as we saw in part (b)) of the most recent example are used frequently in topographic maps of mountainous regions and are called contour maps.

Question to the Class: What does the graph of $f(x, y) = x + y - 2$ look like?

Definition: A function of n variables is a rule that assigns a number $z = f(x_1, x_2, \dots, x_n)$ to an n -tuple $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Remark: If $n \geq 3$ functions of n variables can be very difficult or impossible to picture visually. One simple example of how one might picture a function of 3 variables is representing how a surface changes over time.

For example, consider the function $f(x, y, z) = x^2 + y^2 + z^2$. If we think of $f(x, y, z)$ as being a time variable, the surface is a sphere of radius one when time is one, a sphere of radius 2 when time is 4, etc
(Note: This is certainly not the only way one might interpret this function).

Example 4: At a bakery there are n ingredients used to make all possible items. Suppose that x_i represents the number of units of the i^{th} ingredient used by the bakery on a particular day. Suppose also that the i^{th} ingredient costs c_i per unit.

a) If $C(x_1, x_2, \dots, x_n)$ is the daily expense function (on ingredients) for the bakery, find a formula for $C(x_1, x_2, \dots, x_n)$.

b) Write the domain of the function in (a) in set notation.